

Newton's Rings

1 Aim

To determine wavelength of Sodium light by measuring diameter of Newton's rings.

2 Introduction

This experiment demonstrated interference by division of amplitude (amplitude splitting). Newton's rings are circular interference fringes produced by a very thin film between a glass plate and a lens with a large radius of curvature. Light from a Sodium lamp is rendered parallel by a convex lens. These rays fall on a glass plate which is inclined at an angle of 45° with respect to vertical and is reflected on to the lens. The interference O's see is due to two beams of light, one reflected from the top of the glass plate and the other from the lower surface of the thin film of air.

The wavelength of (almost) monochromatic light is given by

$$\lambda = \frac{D_n^2 - D_m^2}{4R(n - m)},$$

where R is the radius of curvature of the surface of the lens and D_n and D_m are the diameters of n th and m th ring (dark or bright).

3 Apparatus in the lab and procedure

1. Clean the surface of the glass plate and the lens and place the lens with curved surface on the glass plate.
2. Adjust the microscope such that the point of intersection of cross wires coincide with the centre of the ring system and one of the cross wires is perpendicular to the scale.



Figure 1: Newton's ring apparatus

3. Slide the microscope to one side till the cross wire is at the outer edge of one of the rings (say 20th) and move the microscope to the inner edge of the fringe on the other side. Repeat the process by starting from the inner edge (this will reduce the uncertainty in determining the centre of the fringe).
4. Repeat and take many readings for different rings.
5. Plot a graph between D_n^2 and the number of the rings n . Measure the slope of the best fit line and find the wavelength λ from the slope and the radius of curvature.

4 Precautions

1. The Sodium lamp should always be plugged into its power supply and not in the direct power line.
2. The surface of the lens and the surface of the glass plate should be cleaned thoroughly before starting the experiment.
3. The micrometer screw should be moved only in one direction to avoid back-lash error.

5 Points to Ponder

1. If instead of air, we had an oil film, how will the results change.
2. Using the wavelength of the monochromatic light as a standard, determine the radius of curvature of the given lens.